PRE-ALGEBRA (7TH) · UNIT 2: SOLVING ONE-STEP EQUATIONS AND INEQUALITIES

Movie Night Fundraiser

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students run a school movie night. They simplify cost and profit expressions with properties, solve one-step equations to find sales, and use one-step inequalities for the profit goal and the gym capacity.

The combination of a goal (at least) and a rule (at most) shows why inequalities matter.

Standards Addressed

- 7.EE.A.1 Apply properties to generate equivalent expressions
- 7.EE.B.3 Solve problems with rational numbers
- 7.EE.B.4 Write and solve equations and inequalities
- 6.EE.B.7 One-step equations

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: How many tickets do we need to sell? Part 1 simplifying expressions.
Day 2	Part 2 one-step equations.
Day 3	Part 3 inequalities and the fundraiser plan.

Materials

- Student handout and Price Board
- Calculators
- Poster or slide template

How the Three Versions Differ

Level	What changes
CP2	Distributive property with numbers, combining like terms, whole-number equations, and inequalities with starters.
CP1	Profit expressions, decimal equations, dividing by a negative, and a constrained plan.
Honors	Proofs of equivalence, rational coefficients, attendance model with two constraints, and ticket price optimization.

Watch For (Common Misconceptions)

- Distributing to only the first term.
- Forgetting to flip the inequality when dividing by a negative.
- Using revenue instead of profit.

Extension Ideas

- Use a spreadsheet to test different prices.
- Survey students about snack prices.
- Plan a second event with different costs.

Answer Key – CP2

Part 1: Simplify the Costs

#	Task	Answer
1	Use the distributive property: a family buys 4 tickets and 4 popcorns. Show 4(6 + 3) two ways.	4(9) = 36 and 4(6) + 4(3) = 24 + 12 = 36
2	Simplify 6t + 3p + 2t + p.	8t + 4p
3	Which property is shown? 3 + 6 = 6 + 3	Commutative Property of Addition
4	Simplify 5(2x + 4).	10x + 20

Part 2: Equations

#	Task	Answer
5	Ticket sales were $540. How many tickets were sold?	6t = 540 → t = 90
6	Popcorn sales were $186. How many bags were sold?	3p = 186 → p = 62
7	After paying for the projector, the council had $415 left. How much did they collect?	x − 75 = 415 → x = 490
8	Check one answer by substituting it back into the equation.	e.g., 6(90) = 540 ✓

Part 3: Inequalities

#	Task	Answer
9	At most 240 people can be in the gym. 85 people are inside. How many more can come in?	85 + n ≤ 240 → n ≤ 155
10	The council wants at least $300 from ticket sales. How many tickets t must they sell?	6t ≥ 300 → t ≥ 50
11	Write your own inequality about drinks and solve it.	Answers vary; e.g., 2d ≥ 100 → d ≥ 50.

Reflection

#	Prompt	Look for
R1	How do inverse operations help you solve an equation?	They undo the operation on the variable so it is alone, keeping both sides equal.
R2	How is the answer to an inequality different from an equation?	An inequality has many solutions shown as a ray on a number line.

Answer Key – CP1

Part 1: Profit Expressions

#	Task	Answer
1	Write an expression for the profit on p popcorns and d drinks (price minus our cost).	$(3 - 1)p + (2 - 0.50)d = 2p + 1.5d$
2	Write and simplify an expression for total profit including tickets and the projector.	$6t + 2p + 1.5d - 75$
3	A "combo" is 1 ticket, 1 popcorn, and 1 drink for $10. Simplify the profit on c combos.	$10c - (1 + 0.50)c = 8.5c$
4	Simplify $-3(2x - 5) + 4x$ and name the properties you used.	$-6x + 15 + 4x = -2x + 15$; distributive, commutative, combining like terms

Part 2: Equations

#	Task	Answer
5	The combo profit was $595. How many combos were sold?	$8.5c = 595 \rightarrow c = 70$
6	Drink profit was $93. How many drinks were sold?	$1.5d = 93 \rightarrow d = 62$
7	The temperature in the gym fell 6 degrees to 64°F. What was it before?	$x - 6 = 64 \rightarrow x = 70$°F
8	The council split the profit equally among 4 teams; each got $287.50. What was the total profit?	$\frac{P}{4} = 287.50 \rightarrow P = \$1{,}150$

Part 3: Inequalities & Plan

#	Task	Answer
9	The council already has $650 profit. How much more profit p is needed? Solve and graph.	$650 + p \geq 1{,}200 \rightarrow p \geq 550$
10	The air conditioner lowers the gym temperature 2 degrees per hour, and the total change must be no lower than −8 degrees. Solve $-2h \geq -8$ and explain what the answer means.	Divide by −2 and flip: $h \leq 4$; the cooling can run at most 4 hours.
11	Make a plan for selling tickets, combos, and snacks that reaches the $1,200 goal without going over 240 people. Show your calculations.	Answers vary; check profit ≥ 1,200 and people ≤ 240.

Reflection

#	Prompt	Look for
R1	Why do you reverse the inequality symbol when you divide by a negative?	Multiplying or dividing by a negative reverses order on the number line (e.g., $2 < 3$ but $-2 > -3$).
R2	How did simplifying expressions make the plan easier?	Fewer terms made it faster to compute profit for many choices.

Answer Key – Honors

Part 1: Expression Analysis

#	Task	Answer
1	A family pack is 4 tickets, 3 popcorns, and 4 drinks for $38. Write the profit per pack and simplify. Is the pack a better deal for the council than selling the items separately?	Pack profit: 38 − 3(1) − 4(0.50) = $33. Separately: 24 + 6 + 6 = $36 profit. Separate sales earn $3 more.
2	Prove 6(t + 2) − 2(3t − 1) is the same for every t. What is its value?	6t + 12 − 6t + 2 = 14 for every t
3	Simplify $\frac{3}{4}(8x - 12) - \frac{1}{2}(4x + 6)$.	6x − 9 − 2x − 3 = 4x − 12
4	Write two different expressions that equal 12n − 18 and explain the property used.	e.g., 6(2n − 3) (distributive), 3(4n − 6), 12n + (−18)

Part 2: Equations with Rational Numbers

#	Task	Answer
5	Candy profit is $0.75 per piece. Profit was $67.50. How many pieces sold?	0.75c = 67.50 → c = 90
6	$\frac{2}{3}$ of the seats were filled with 160 people. How many seats are there?	$\frac{2}{3}$s = 160 → s = 240
7	After a −$18.25 refund, the balance was $1,043.75. What was it before?	x + (−18.25) = 1,043.75 → x = $1,062
8	$-\frac{5}{4}x = 35$. Solve and write a movie-night story that this equation could describe.	x = −28; e.g., a cooling change or a loss scenario.

Part 3: Constraints & Optimization

#	Task	Answer
9	Each person buys one ticket and spends on average $3.50 on snacks with average profit $2.10. Write and solve an inequality for the number of people needed to reach the goal.	Profit per person 6 + 2.10 = 8.10; 8.10n − 75 ≥ 1,200 → 8.10n ≥ 1,275 → n ≥ 157.4 → 158 people (one-step after adding 75)
10	Graph the possible numbers of people on a number line (use both the goal and the gym rule).	158 ≤ n ≤ 240
11	Recommend a ticket price change that would reach the goal with only 130 people. Justify with an inequality.	(x + 2.10)(130) − 75 ≥ 1,200 → x + 2.10 ≥ 9.81 → x ≥ $7.71, so about $7.75 per ticket.

Reflection

#	Prompt	Look for
R1	When does a "deal" earn less profit for the seller? Use your family pack result.	When the pack price is less than the separate prices while costs stay the same, profit drops.
R2	How do two inequalities together describe the possible attendance?	The goal gives a minimum and the gym rule gives a maximum; the overlap is the possible range.